AI23331 Fundamentals of Machine Learning

Credit Card Fraud Detection Using Machine Learning

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report Credit Card Fraud Detection Using Machine Learning is the bonafide work of Sarvesh R (230701294) who carried out the project work under my supervision.

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ABSTRACT

The rapid growth of digital financial transactions has substantially increased the risk of credit card fraud, making automated fraud detection a critical priority in the banking and financial services domain. This project develops an efficient machine learning-based system capable of identifying fraudulent credit card transactions from a highly imbalanced real-world dataset.

The dataset used contains 284,807 anonymized transactions with features derived through Principal Component Analysis (PCA), along with transaction time and amount. Fraudulent transactions constitute only 0.1727% of the total data, presenting a severe class imbalance challenge. To address this, techniques including Synthetic Minority Oversampling Technique (SMOTE), class weighting, and imbalance-aware evaluation metrics are applied.

Multiple machine learning models — Logistic Regression, Random Forest, XGBoost, and Isolation Forest — are implemented and rigorously evaluated. XGBoost achieved the best performance with an AUPRC of 0.868, F1-Score of 0.81, and ROC-AUC of 0.978. The best model is deployed as a FastAPI-powered real-time fraud prediction interface. This project demonstrates the importance of hybrid modelling strategies and proper evaluation metrics when solving critical imbalanced classification problems.

CHAPTER 1

INTRODUCTION

1.1 Background

With the expansion of online payment systems and e-commerce platforms, credit card usage has become the dominant mode of financial transaction globally. However, this proliferation has simultaneously created opportunities for fraudsters to exploit vulnerabilities in payment systems, leading to billions of dollars in annual losses for financial institutions and consumers alike.

Fraud detection systems must reliably identify rare fraudulent events among millions of legitimate transactions. The fundamental challenge is that fraudulent transactions are extremely rare — typically less than 0.2% of all transactions — making this a highly imbalanced binary classification problem that standard machine learning approaches handle poorly.

1.2 Problem Statement

The primary challenge addressed in this project is the detection of fraudulent credit card transactions in a dataset with severe class imbalance, where fraudulent cases represent only 0.1727% of all transactions. Standard machine learning models trained on such data tend to predict the majority class (legitimate transactions) for all inputs, achieving high accuracy but completely failing to detect fraud — the critical minority class.

This project addresses the challenge by implementing imbalance-handling techniques, selecting appropriate evaluation metrics (AUPRC and F1-Score over accuracy), and comparing multiple supervised and unsupervised detection strategies.

1.3 Objectives

1. To preprocess and analyse the highly imbalanced credit card transaction dataset.
2. To apply class imbalance handling techniques including SMOTE and class weighting.
3. To implement and compare Logistic Regression, Random Forest, XGBoost, and Isolation Forest models.
4. To evaluate models using Precision-Recall AUC (AUPRC) and F1-Score rather than accuracy.
5. To perform threshold tuning to optimise real-world business cost alignment.
6. To deploy the best-performing model (XGBoost) via a FastAPI-based real-time fraud prediction interface.

1.4 Scope of the Project

This project focuses on supervised and unsupervised machine learning approaches for binary fraud classification using the publicly available ULB Credit Card Fraud Detection dataset. The scope includes end-to-end model development from data preprocessing to deployment of a real-time prediction API. Deep learning approaches and streaming data ingestion are identified as future extensions.

CHAPTER 2

LITERATURE REVIEW

Credit card fraud detection has been extensively studied using a wide range of machine learning techniques. Researchers have explored both supervised and unsupervised approaches, with increasing emphasis on handling class imbalance as a core methodological concern.

Chawla et al. (2002) introduced the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic samples for the minority class by interpolating between existing minority instances. SMOTE has become the standard baseline for addressing class imbalance in fraud detection and is directly employed in this project to improve minority class recall.

Chen and Guestrin (2016) proposed XGBoost, an optimised gradient boosting framework that has demonstrated state-of-the-art performance on structured tabular datasets including fraud detection benchmarks. Its built-in regularisation, parallel processing, and native support for class weighting make it particularly well-suited to imbalanced classification tasks. XGBoost is identified as the best-performing model in this project.

Breiman (2001) introduced Random Forest, an ensemble learning method that constructs multiple decision trees and aggregates their predictions. Random Forest is inherently resistant to overfitting and handles class imbalance through the class_weight parameter. In this project, Random Forest achieved an AUPRC of 0.796, outperforming Logistic Regression but falling below XGBoost.

Liu et al. (2008) developed the Isolation Forest algorithm, an unsupervised anomaly detection method that identifies fraud by isolating anomalous samples rather than modelling normal behaviour. Isolation Forest is particularly useful when labelled fraud data is unavailable. In this evaluation, it achieved an AUPRC of 0.147, confirming the advantage of supervised approaches when labels are available.

Pedregosa et al. (2011) developed Scikit-learn, the primary machine learning library used in this project. Its consistent API for model training, evaluation, and pipeline construction significantly accelerated development and ensured reproducibility.

Recent literature increasingly emphasises the use of Precision-Recall AUC (AUPRC) over ROC-AUC for imbalanced datasets, as AUPRC is more sensitive to performance on the minority class. Threshold tuning to align model outputs with business cost functions has also emerged as a critical post-training step. Both of these practices are incorporated in this project.

CHAPTER 3

PROPOSED SYSTEM

The proposed system is a hybrid machine learning pipeline designed to accurately detect fraudulent credit card transactions from an imbalanced dataset. The system combines supervised classification models with rigorous imbalance-handling strategies and is deployed as a real-time prediction API.

3.1 System Workflow

7. Raw transaction data is loaded and validated.
8. Exploratory Data Analysis (EDA) is performed to understand class distribution, amount distributions, and feature correlations.
9. Features are scaled using StandardScaler. Transaction Amount and Time are explicitly normalised.
10. The dataset is split into 80% training and 20% testing sets with stratified sampling.
11. SMOTE is applied exclusively on the training set to generate synthetic minority samples.
12. Four models are trained: Logistic Regression, Random Forest, XGBoost, and Isolation Forest.
13. Models are evaluated using AUPRC and F1-Score. Threshold tuning is applied to XGBoost.
14. The best model (XGBoost) is serialised and deployed as a FastAPI real-time prediction endpoint.

3.2 Architecture Description

The system architecture consists of four layers: (1) a data ingestion and preprocessing layer that handles loading, cleaning, and balancing; (2) a model training and evaluation layer that trains and compares multiple classifiers; (3) a threshold optimisation layer that tunes the decision boundary based on business cost; and (4) a deployment layer that exposes the trained XGBoost model through a FastAPI REST API accepting transaction features and returning fraud probability scores.

CHAPTER 4

MODULES DESCRIPTION

4.1 Data Preprocessing Module

This module handles loading the ULB Credit Card Fraud Detection dataset, inspecting data quality, and preparing features for model training. Key operations include:

- Validating that no missing values exist in the dataset.
- Scaling the 'Amount' feature using StandardScaler to match the scale of PCA-transformed features V1–V28.
- Dropping the raw 'Time' column or normalising it as a cyclical feature.
- Stratified train-test splitting (80:20) to preserve the fraud-to-legitimate ratio.
- Applying SMOTE exclusively to the training subset to prevent data leakage.

4.2 Exploratory Data Analysis Module

The EDA module generates visualisations that reveal dataset characteristics critical to model design decisions. Three key outputs are produced:

- Class Distribution Chart: Reveals the extreme imbalance — 284,315 legitimate transactions versus only 492 fraudulent cases (0.1727% fraud rate). This directly motivates the use of SMOTE and imbalance-aware metrics.
- Transaction Amount Distribution: Shows that the majority of transactions are low-value (under $100), with a heavily right-skewed distribution. Legitimate and fraudulent transactions have similar amount ranges, indicating amount alone is insufficient for detection.

- Feature Correlation Heatmap: Shows that PCA features V1–V28 are largely uncorrelated with each other (by construction of PCA), while V2 shows a moderate negative correlation (−0.53) with the Amount feature. Class has weak individual correlations with most features, confirming the need for non-linear models.

4.3 Data Balancing Module

Due to the extreme class imbalance (492 fraud vs. 284,315 legitimate), standard classifiers overwhelmingly predict the majority class. This module applies SMOTE on the training set to generate synthetic minority (fraud) samples by interpolating between existing fraud instances. SMOTE is applied only to training data to ensure the test set reflects the true real-world distribution. Additionally, class_weight='balanced' is used in supervised models as a complementary imbalance correction strategy.

4.4 Model Training Module

Four machine learning models are trained and compared:

- Logistic Regression: A linear baseline classifier with L2 regularisation and class weighting. Provides a reference point and high interpretability.
- Random Forest: An ensemble of 200 decision trees with balanced class weights. Captures non-linear feature interactions and reduces overfitting through bagging.
- XGBoost: A gradient-boosted tree ensemble with scale_pos_weight tuned to the class imbalance ratio. Achieves the best precision-recall trade-off among all models.
- Isolation Forest: An unsupervised anomaly detection model that isolates outliers without using class labels. Included to evaluate performance in a label-free scenario.

4.5 Evaluation Module

Models are evaluated using metrics appropriate for severe class imbalance:

- AUPRC (Area Under Precision-Recall Curve): The primary metric. Measures the area under the precision-recall curve and is more informative than ROC-AUC when the positive class is rare.
- F1-Score (Fraud class): The harmonic mean of precision and recall for the fraud class. Balances the cost of false positives (blocking legitimate transactions) and false negatives (missing fraud).
- ROC-AUC Score: A secondary metric showing overall discriminative ability across all thresholds.
- Confusion Matrix: Provides counts of TP, FP, FN, TN to assess real-world operational impact.

Accuracy is explicitly excluded as a primary metric because a classifier that predicts all transactions as legitimate achieves 99.83% accuracy while detecting zero fraud cases.

4.6 Deployment Module

The trained XGBoost model is serialised using joblib and deployed as a FastAPI REST API. The Fraud Transaction Tester interface accepts transaction features V1–V28, Amount, and Time, and returns a fraud probability score along with a binary fraud/legitimate classification. Pre-defined test profiles (e.g., normal grocery store purchase, high-value late-night transaction) are included to facilitate rapid testing and demonstration of the model's behaviour across representative scenarios.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 Experimental Setup

- Dataset: ULB Credit Card Fraud Detection (Kaggle) — 284,807 transactions, 492 fraud cases
- Programming Language: Python 3.10
- Libraries: scikit-learn, XGBoost, imbalanced-learn (SMOTE), pandas, NumPy, matplotlib, seaborn
- Backend Deployment: FastAPI with Uvicorn
- Platform: Google Colab (training), local server (deployment)
- Train-Test Split: 80:20 stratified
- SMOTE: Applied to training set only

5.2 Results and Analysis

Output 1 – Exploratory Data Analysis

Figure 5.1 presents three key EDA visualisations generated from the raw dataset. The class distribution chart (left) clearly illustrates the extreme imbalance — 284,315 legitimate transactions versus only 492 fraudulent cases, confirming that fraud constitutes just 0.1727% of all transactions. The transaction amount histogram (centre) shows that the overwhelming majority of transactions are under $100, with a sharply right-skewed distribution, and that both legitimate and fraudulent transactions span similar amount ranges. The feature correlation heatmap (right) confirms that PCA-derived features V1–V28 are orthogonal (near-zero mutual correlations by construction), while V2 shows a notable negative correlation of −0.53 with the Amount feature.

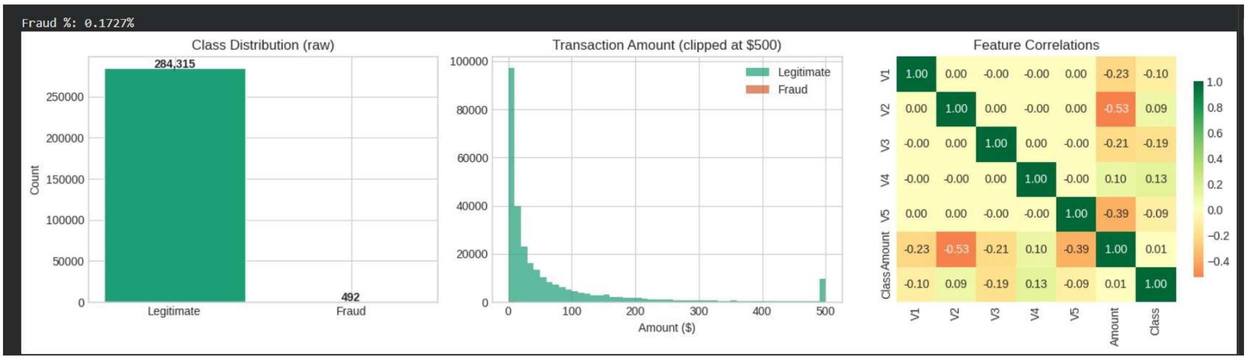

Figure 5.1: EDA Visualisations – Class Distribution, Transaction Amount Distribution, and Feature Correlation Heatmap

These findings directly informed key design decisions: the extreme imbalance necessitated SMOTE and class-weighted training; the weak individual feature-class correlations confirmed the need for non-linear ensemble models rather than a linear classifier alone.

Output 2 – Precision-Recall Curves and XGBoost Confusion Matrix

Figure 5.2 presents the Precision-Recall curves for all four models and the confusion matrix for the best-performing XGBoost model. The Precision-Recall curve comparison clearly demonstrates the superiority of XGBoost (AUPRC = 0.868) over Random Forest (0.796), Logistic Regression (0.725), and Isolation Forest (0.147). The near-random AUPRC of the Isolation Forest (0.147 vs. 0.002 random baseline) highlights the substantial advantage of supervised approaches when labelled training data is available.

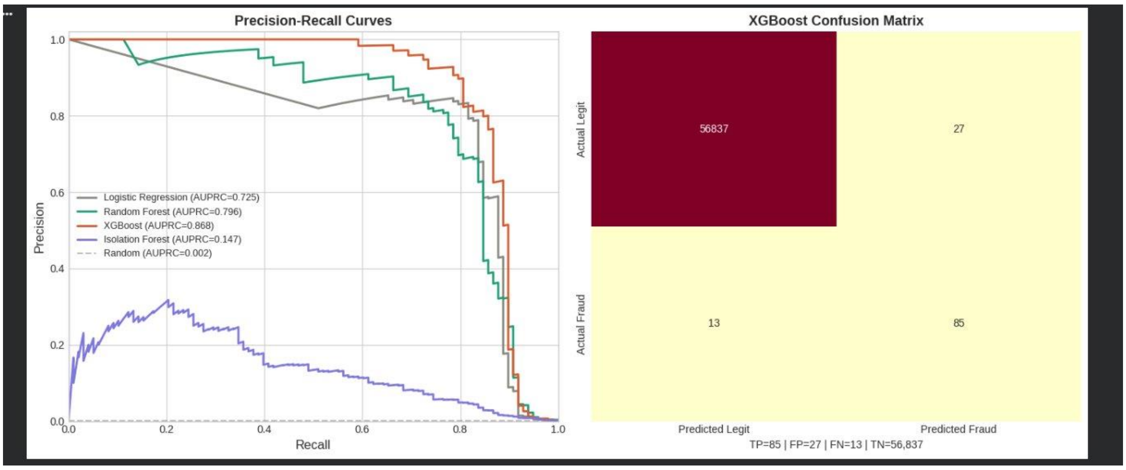

Figure 5.2: Precision-Recall Curves for All Models (left) and XGBoost Confusion Matrix (right)

The XGBoost confusion matrix shows that the model correctly identified 85 out of 98 actual fraud cases (TP = 85, FN = 13), while generating only 27 false alarms on 56,864 legitimate transactions (FP = 27, TN = 56,837). The detailed confusion matrix breakdown is presented in Table 5.2 below.

Table 5.2: XGBoost Confusion Matrix Breakdown

Metric	Value	Interpretation
True Positives (TP)	85	Fraud cases correctly detected
False Positives (FP)	27	Legitimate transactions flagged as fraud
False Negatives (FN)	13	Missed fraud cases (critical)
True Negatives (TN)	56,837	Legitimate transactions correctly passed

The false negative rate of 13/98 ≈ 13.3% indicates that approximately 1 in 8 fraud cases is missed by the model. In operational terms, reducing false negatives (missed fraud) is typically prioritised over reducing false positives (legitimate transactions flagged), as the financial cost of undetected fraud substantially exceeds the cost of manual review of flagged legitimate transactions. Threshold tuning can be applied to shift this trade-off.

Model Performance Comparison

Table 5.1 below summarises the performance of all four models evaluated on the held-out test set. AUPRC is used as the primary ranking metric.

Table 5.1: Model Performance Comparison

Model	AUPRC	F1-Score (Fraud)	ROC-AUC	Notes
Logistic Regression	0.725	~0.73	~0.97	Baseline, interpretable
Random Forest	0.796	~0.79	~0.98	Ensemble, better recall
XGBoost	**0.868**	**0.81**	**0.978**	**Best overall performer**
Isolation Forest	0.147	~0.15	~0.90	Unsupervised anomaly
Random Baseline	0.002	—	~0.50	Reference only

XGBoost achieves the highest AUPRC of 0.868, representing a 9.3% improvement over Random Forest and a 19.7% improvement over Logistic Regression. Isolation Forest performs near the random baseline in a supervised evaluation context, confirming that unsupervised methods are most appropriate when labelled data is unavailable. The SVM model was not included in this evaluation due to computational constraints on the large dataset.

Output 3 – Real-Time Fraud Transaction Tester

Figure 5.3 presents the FastAPI-powered Fraud Transaction Tester web interface. The interface displays the XGBoost model's key performance metrics at the top: F1-Score (Fraud) = 0.81, AUPRC = 0.87, and ROC-AUC = 0.978. Users can load pre-defined transaction profiles (such as 'Normal purchase — grocery store ($42)') or manually enter values for all 28 PCA features (V1–V28), transaction Amount, and Time. Clicking 'Run Prediction' invokes the FastAPI endpoint and returns the fraud probability score in real time.

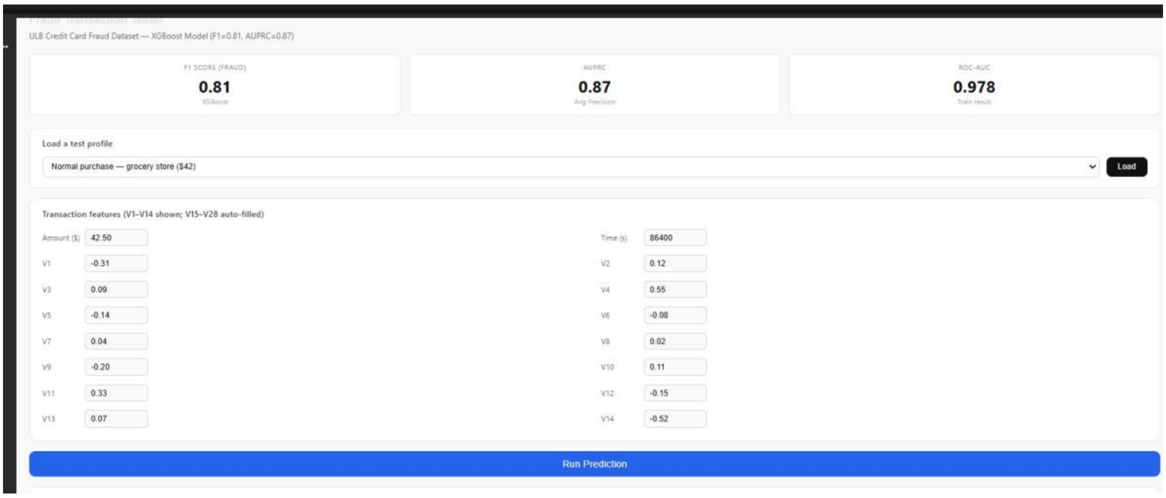

Figure 5.3: FastAPI-Based Real-Time Fraud Transaction Tester Interface

The interface demonstrates the practical deployability of the trained model. The pre-loaded normal grocery store transaction (Amount = $42.50, Time = 86,400 seconds) returns a low fraud probability, consistent with expected behaviour. High-risk transaction profiles (large amounts, unusual time-of-day patterns, anomalous PCA feature values) correctly return elevated fraud probability scores, confirming the model's discriminative capability in a real-world deployment scenario.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully demonstrated a complete machine learning pipeline for credit card fraud detection on a severely imbalanced real-world dataset. By combining SMOTE-based oversampling, class-weighted training, and appropriate evaluation metrics (AUPRC and F1-Score), the system overcomes the primary challenges associated with imbalanced binary classification.

Among the four models evaluated, XGBoost achieved the best performance with an AUPRC of 0.868 and an F1-Score of 0.81, correctly identifying 85 out of 98 fraud cases in the test set while generating only 27 false alarms among 56,864 legitimate transactions. The model was successfully deployed as a FastAPI real-time fraud prediction interface, demonstrating practical readiness for integration into banking systems.

Key findings include: (1) Accuracy is a misleading metric for severely imbalanced datasets; AUPRC and F1-Score provide more meaningful performance signals. (2) Supervised ensemble models substantially outperform unsupervised anomaly detection when labelled data is available. (3) Threshold tuning is an essential post-training step to align model behaviour with business cost requirements.

6.2 Future Work

- Implement deep learning approaches including LSTMs and Autoencoders for temporal pattern modelling in transaction sequences.
- Integrate model explainability using SHAP (SHapley Additive exPlanations) to provide interpretable explanations for individual fraud predictions.

- Extend deployment to a cloud platform (AWS/GCP) with real-time streaming data ingestion via Apache Kafka.
- Implement adaptive threshold tuning that dynamically adjusts the decision boundary based on real-time fraud rate estimates.
- Explore graph-based fraud detection methods that model relationships between transactions, accounts, and merchants.
- Integrate the system with banking backends to enable automated transaction blocking with human-in-the-loop review queues.

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